Multimodal Misinformation Detector Agent

Data Science and AI Lab Project

Milestone 6 Report

Group 9

Rajya Lakshmi Jampani	22f3002986@ds.study.iitm.ac.in
Faiz Malik	23f2004839@ds.study.iitm.ac.in
Sachin Singh	21f1003251@ds.study.iitm.ac.in
G. Nikhil Chand	21f1003825@ds.study.iitm.ac.in
C. Glenn Varshan	22f2000961@ds.study.iitm.ac.in

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1. Objective

This milestone focuses on the end-to-end deployment, documentation, and consolidation of the multimodal misinformation detection system. Building upon the model development, experimentation, and evaluation completed in earlier milestones, Milestone 6 transitions the project from research to a fully deployable and well-documented solution.

The objective is to package the trained multimodal classifiers into a production-ready system, prepare technical and user-facing documentation, and finalize the project with conclusions, future directions, references, and appendices. This milestone also includes the creation of sample inputs and outputs demonstrating how the deployed system behaves in real-world usage.

2. Deployment and Documentation

2.1 Deployment Platform

- **Frontend**: Gradio interface (two tabs for text and image detection).
- **Backend**: Python-based application hosted on Hugging Face Spaces.
- **Model Hosting**: All models loaded dynamically from Hugging Face Hub.
- **LLM Integration**: Uses Hugging Face Inference API for explanation generation.

2.2 Inference Pipeline

- User provides text claim or image.
- If text: evidence retrieval + DeBERTa classification + Llama explanation.
- If image: processed by CLIP vision encoder + fine-tuned classifier head.
- Results displayed in Gradio markdown output.

2.3 Example Outputs

Multimodal Misinformation Detector

Text Detector | Image Detector

Enter Claim

Sun rises in the west

Classify Claim

Prediction: The claim is Fake.

Explanation: The Sun appears to rise in the east and set in the west due to Earth's rotation. This is a widely observed phenomenon and a fundamental aspect of our planet's geography. The statement that the Sun rises in the west contradicts this established fact.

Confidence: High.

Multimodal Misinformation Detector

Text Detector | Image Detector

Enter Claim

India is in Asian continent

Classify Claim

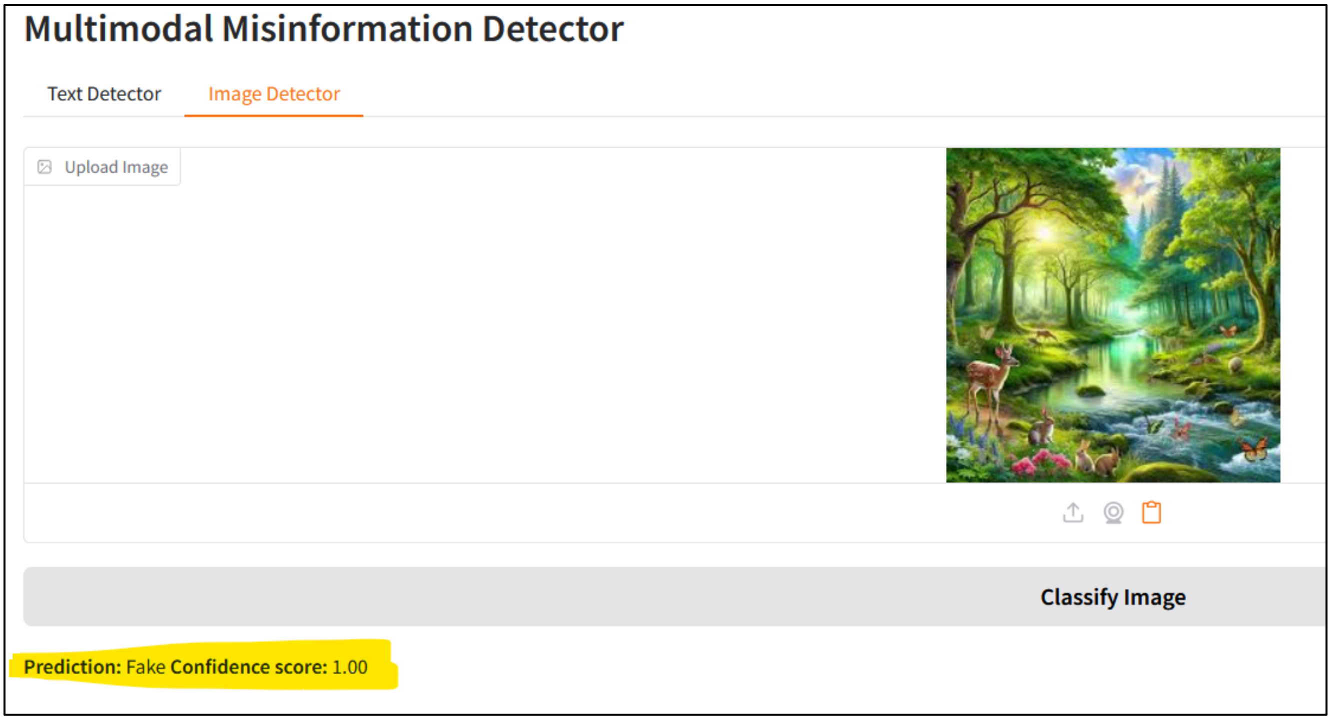
Prediction: The claim is Real.

Explanation: India is indeed located within the Asian continent, sharing borders with countries like China and Nepal. The Indian subcontinent is a distinct geographic region that is part of the larger Asian continent. This is due to its geographical features, such as the Himalayan Mountains, and its position on the Indian Tectonic Plate.

Confidence: High.

Classify Image

Prediction: Real **Confidence score:** 0.49



2.4 Documentation

Full setup, dependency, and model usage instructions are included in *README.md* and technical documentation accessible at Github Repo. All modules are modular and reusable for future extensions.

3. Conclusion and Future Work

This project demonstrates an end-to-end multimodal misinformation detection pipeline integrating retrieval, classification, and explainability. The combination of transformer-based NLP models and CLIP-based vision models enables more robust multimodal verification than unimodal approaches. The deployed Hugging Face Space provides a functional and user-friendly interface accessible to both technical and non-technical users.

Future Directions:

- Experiment with larger LLMs or multimodal transformers.
- Extend to video or audio misinformation detection.

- Develop REST API endpoints for programmatic access.
- Integrate user feedback to improve model accuracy over time.
- Implement real-time monitoring of usage and performance metrics.
- Set up unit and integration tests for core functions and models.

4. References

- Thorne et al., “FEVER: A Large-scale Dataset for Fact Extraction and Verification,” EMNLP 2018.
- He et al., “DeBERTa: Decoding-enhanced BERT with Disentangled Attention,” ICLR 2021.
- Radford et al., “Learning Transferable Visual Models from Natural Language Supervision,” ICML 2021.
- OpenAI, “CLIP Model Card,” 2021.
- Meta AI, “Llama 3.1 Technical Report,” 2024.

5. Appendix

5.1 Training parameters

- **Text Classifier**: Learning Rate = 2e-5, Epochs = 3, Batch Size = 32.
- **Image Classifier**: Learning Rate = 1e-4, Epochs = 3, Batch Size = 32.

5.2 API References

- Google Fact Check API
- Tavily Search API
- Wikipedia API
- Hugging Face Inference API

5.3 Deployment Link

https://huggingface.co/spaces/group9-dsailab/multimodal_misinfo_detector

5.4 License Information

- **Code**: MIT License
- **Datasets**: CC BY-SA 4.0 (FEVER), CC BY-SA 3.0 (Wikipedia)
- **Models**: Licensed under respective Hugging Face terms